



PVINSIGHT

Photovoltaic Energy Forecasting System for Renewable Energy Management

Deep-learning forecasting of solar irradiance & module temperature for accurate PV energy estimation at the Azraq Solar Plant

Zaina Ibrahim · Wasfi Al Qudah · Abdullah Aldahleh

Supervisor: Ms. Wasan Almaitaah

PRINCESS SUMAYA UNIVERSITY FOR TECHNOLOGY · KING HUSSEIN FACULTY OF COMPUTING SCIENCES

Department of Data Science and AI

Artificial Intelligence Track

PV Renewable Energy

1 INTRODUCTION

Solar energy generation is **highly dependent on weather conditions**, making output variability one of the biggest challenges in modern grid operations.

Accurate short-term forecasting of irradiance and module temperature directly improves **renewable energy management**, dispatch scheduling, and operational planning for PV plants.

PVInsight delivers a complete data-to-dashboard pipeline that predicts and visualizes solar power output to support smarter, data-driven energy decisions.



2 PROJECT OBJECTIVES

- Predict** solar irradiance and average module temperature from historical PV plant data.
- Forecast** future irradiance and module temperature using LSTM deep-learning models.
- Estimate** photovoltaic energy generation through a physics-based PV energy equation.
- Visualize** predictions and KPIs through an interactive Streamlit dashboard.

3 DATASET & PREPROCESSING

Operational data from the **Azraq Solar Power Plant** in Jordan, cleaned, resampled and aligned for time-series modeling.

2020
– 2022
3 YEARS

15
min
SAMPLING INTERVAL

Azraq
SOLAR PLANT
JORDAN

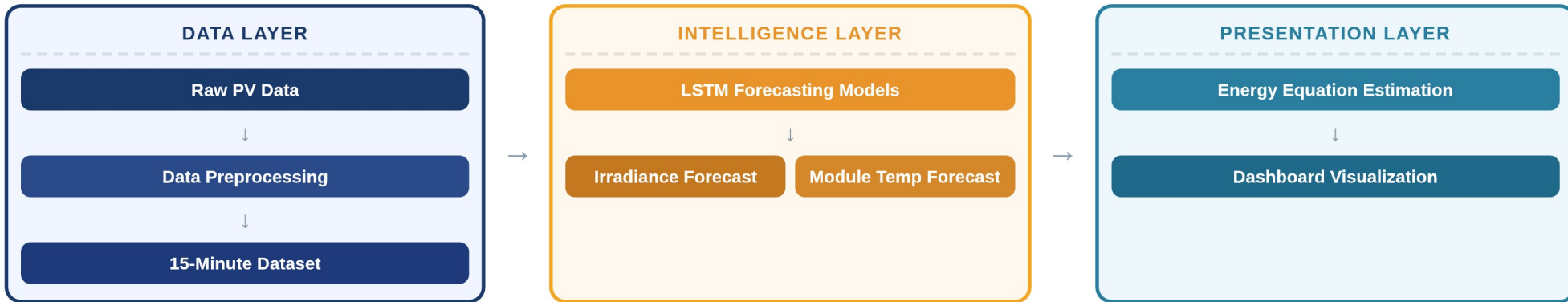
6 ENERGY ESTIMATION

$$E = G \cdot \eta_{ref} \cdot [1 + \gamma(T - 25)]$$

E Estimated PV energy output
T Module temperature (°C)
γ Temperature coefficient of power

G Solar irradiance (W/m²)
η_{ref} Reference panel efficiency
25 Reference temperature (°C, STC)

4 SYSTEM ARCHITECTURE



5 MACHINE LEARNING MODELS

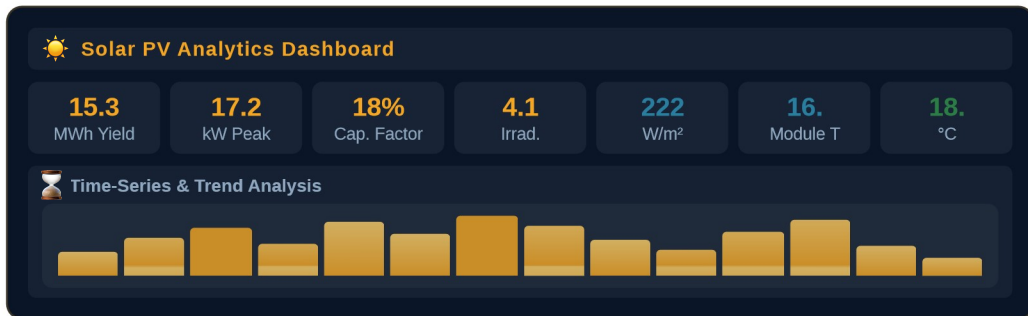
Two parallel **Long Short-Term Memory (LSTM)** networks were trained on the 15-minute Azraq dataset to capture sequential dependencies in the weather and PV signals.

MODEL	PURPOSE
LSTM 1	Solar Irradiance Forecasting
LSTM 2	Module Temperature Forecasting

7 TESTING RESULTS

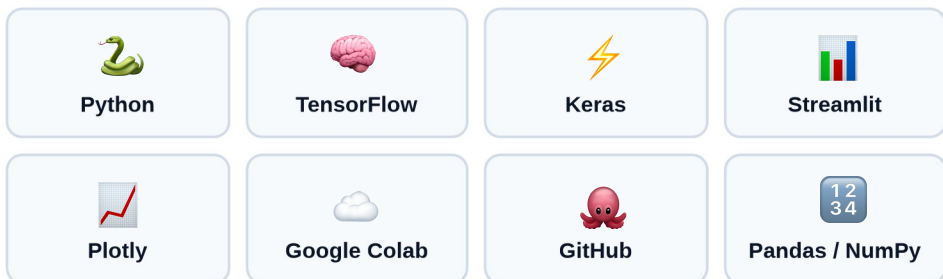
LSTM · Module Temperature		LSTM · Solar Irradiance	
MAE	0.9299 °C	MAE	20.5494 W/m²
RMSE	1.5381 °C	RMSE	38.0835 W/m²
R²	0.9886	R²	0.9870

8 INTERACTIVE DASHBOARD



- KPI tiles** — total yield, peak power, capacity factor, module temperature.
- Time-series & trends** with rolling averages and daily peak markers.
- Distributions, heatmaps** and correlations across irradiance, temperature and output.
- Model-analysis tab** comparing LSTM predictions against measured plant readings.

9 TECHNOLOGIES USED



10 FUTURE WORK

- Real-time forecasting** with live data streams from on-site sensors.
- Cloud deployment** for scalable, multi-plant monitoring and inference.
- Additional weather variables** — humidity, wind speed, cloud cover and aerosol index.
- Advanced hybrid models** combining LSTM with Transformers, CNN-LSTM and physics-informed networks.